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BITE308L: ARTIFICIAL INTELLIGENCE

Adaptive Security Awareness Training System

Using Reinforcement Learning and
LLM-Based Scenario Generation

Submitted by

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1. Abstract

Phishing remains difficult to prevent because it targets human judgement as much as technical weaknesses. Most security-awareness programs rely on a fixed collection of examples and present broadly similar material to every employee. In practice, however, one person may react strongly to urgency while another may be more influenced by authority, invoices, or requests for credentials.

This project proposes an Adaptive Security Awareness Training System that brings together reinforcement learning, large language model (LLM) scenario generation, and lightweight behavior classification in a closed feedback loop. The system treats urgency, authority impersonation, fake invoices, and credential requests as arms in a multi-armed bandit. During each training round, the bandit chooses a tactic using the evidence gathered from earlier interactions.

The chosen tactic, together with the employee's role and context, is then passed to an LLM to create a new simulated phishing scenario. The employee or simulator responds by clicking, ignoring, or reporting the message. A behavior classifier interprets this response, and the reward module converts it into feedback for the bandit. As more interactions are recorded, the system gradually changes which tactics it presents and gives greater attention to the areas that require more training.

The project is limited to a controlled, consent-based simulation; it does not send real phishing emails or collect real credentials. Its performance will be assessed by comparing the adaptive policy with a random tactic-selection baseline. The expected result is a practical prototype that generates varied scenarios, personalizes training, and makes the adaptation process measurable.

2. Problem Statement

Many security-awareness programs follow a fixed sequence of pre-written phishing examples. Although this approach is easy to deploy, it does not learn from the way an individual employee responds.

Susceptibility is rarely uniform. An employee who easily recognizes a fake invoice may still respond to a message that creates urgency or appears to come from a senior manager. A fixed training plan can therefore spend time repeating familiar lessons while giving too little attention to the employee's weaker areas.

Therefore, the problem addressed by this project is:

> *How can an artificial-intelligence-based security awareness training system dynamically identify an employee's comparatively weak social-engineering tactic and adapt subsequent training scenarios using reinforcement learning and LLM-based scenario generation?*

To address this problem, the project links a multi-armed-bandit agent, an LLM scenario generator, a simulated employee model, and a reward mechanism. Together, these components form a loop in which every response influences the scenario selected for a later round.

3. Motivation

The project is motivated by a simple limitation of conventional awareness training: the same lesson does not have the same value for every employee. Phishing decisions are shaped by context, including urgency, authority, financial pressure, and requests for sensitive information. Training is likely to be more useful when it responds to these differences instead of treating every learner in the same way.

Reinforcement learning is well suited to this setting because training takes place over a sequence of decisions. The system chooses a tactic, observes the response, and uses that feedback when choosing the next scenario. Here, the action is the selected social-engineering tactic, the observation is the employee's simulated behavior, and the reward captures the value of that interaction for the training policy.

LLMs complement this decision process by producing fresh scenarios for the selected tactic and employee context. This reduces dependence on a small set of repeated templates. Combining adaptive selection with dynamic generation creates a training experience that can evolve from one round to the next.

4. Literature Survey

Existing phishing research can be grouped into several related areas: technical detection, human susceptibility, awareness training, and adaptive decision-making. Reinforcement learning has mainly been used to detect phishing websites or network intrusions, where an agent learns from repeated feedback (Ariyadasa et al., 2024; Chatterjee & Namin, 2019; Kheddar et al., 2024). Related work has used neural networks and hybrid machine-learning methods to identify phishing emails, malicious URLs, and other harmful online content (Jabbar & Al-Janabi, 2025; Patil & Shekokar, 2025; Smadi et al., 2018).

The growth of large language models has introduced a different challenge. LLMs can produce convincing, context-aware phishing content and can personalize messages at scale (Bethany et al., 2026; Czybik et al., 2026). While this capability increases the threat, it can also be used defensively to create varied and realistic awareness exercises in a controlled environment.

Research on awareness training shows that timing, repeated reminders, clear feedback, and active participation all influence how well users retain what they learn (Madeira & Alves, 2026; Reinheimer et al., 2020; Wen et al., 2019). However, these ideas are often studied separately. Detection systems focus on identifying attacks, training studies focus on educational outcomes, and LLM studies focus on generating or analysing content. This project connects those strands through a policy that selects the next training tactic after observing each response.

5. Research Gap

The literature provides strong solutions for individual parts of the problem, but few systems bring them together in one continuously adapting training process. In particular, there is limited integration of:

1. adaptive reinforcement-learning-based tactic selection;
2. dynamically generated LLM-based training scenarios;
3. employee-specific vulnerability modeling;
4. response-based reward generation; and
5. continuous adaptation of subsequent training rounds.

This gap shifts the focus of the project away from simply deciding whether a message is malicious. Instead, the system asks what type of scenario should be shown next and how the answer should change after each interaction.

6. Objectives

Based on this gap, the project has the following objectives:

1. Model differences in employee susceptibility to several social-engineering tactics within a controlled simulation.
2. Formulate tactic selection as a multi-armed-bandit problem and implement Thompson Sampling as the adaptive policy.
3. Use an LLM to generate varied security-awareness scenarios that reflect the selected tactic and employee role.
4. Interpret simulated employee responses with a lightweight machine-learning classifier.
5. Convert each observed response into feedback that can update the learning policy.
6. Compare the adaptive approach with random tactic selection using consistent synthetic employee profiles.

7. Survey Table

Table 1 summarizes the literature reviewed for the proposed system.

Table 1: Literature Survey and Gap Analysis

Ref.	**Methods**	**Dataset**	**Results Analysis**	**Drawbacks / Observations**
1	Reinforcement learning for phishing detection	Phishing/spoofed website data	RL-based detection performance	Focuses on detection rather than adaptive awareness training
2	Deep reinforcement learning	Phishing website dataset	Evaluates phishing classification	Does not personalize training for employees
3	Neural network with reinforcement learning	Phishing email data	Dynamic phishing email detection	Detection-oriented architecture
4	AI and reinforcement learning	Phishing-related data	Evaluates AI-based detection	Does not provide adaptive employee training
5	Reinforcement-learning-based intrusion detection	Network/security datasets	Reviews RL-based IDS approaches	Focuses on intrusion detection
6	RL + machine learning	Phishing URL data	Evaluates phishing URL detection	Detection rather than awareness training
7	LLM-based phishing generation	Large-scale phishing data	Studies LLM-generated phishing	Focuses on attack generation
8	Personalized phishing using LLMs	Large-scale phishing study	Studies personalized phishing content	Not primarily an adaptive training loop
9	Transformer/LLM-based phishing detection	Phishing datasets	Compares language models	Detection-focused
10	Longitudinal phishing awareness training	Human-participant studies	Studies effectiveness of reminders	Training strategy is not RL-adaptive

Ref.	Methods	Dataset	Results Analysis	Drawbacks / Observations
11	Feedback and engagement strategies	Phishing awareness studies	Analyzes training effectiveness	Does not combine RL and LLM generation
12	Interactive phishing-awareness game	Human participants	Evaluates engagement	Limited adaptive RL-based personalization
13	Psychological predictors of email fraud	Human participant data	Studies susceptibility factors	Does not implement adaptive training
14	Reinforcement learning theory	Simulation environments	Provides RL foundations	General RL framework rather than phishing training
15	Transformer language representation	Large text datasets	Demonstrates contextual language representation	Not specifically an adaptive security-training system

8. Design Methodology

The system is designed as a closed loop so that each training round influences the next one. It begins with a synthetic employee profile containing separate susceptibility scores for urgency, authority, invoice, and credential-request tactics.

Thompson Sampling treats these tactics as the arms of a multi-armed bandit. The agent samples from its current belief about each arm and selects one tactic for the next exercise. The selected tactic and the employee's context are then sent to the LLM, which creates a simulated security-awareness scenario.

After the scenario is presented, the employee simulator produces a response such as clicking, ignoring, or reporting. The behavior classifier processes this response along with the available scenario features. Finally, the reward module returns feedback to the bandit, allowing the agent to update its beliefs before the next round begins.

Figure 1 presents the detailed learning flow, while Figure 2 shows how that flow is integrated with the dashboard, user roles, data storage, and external LLM service.

8.1 Design Methodology Flow

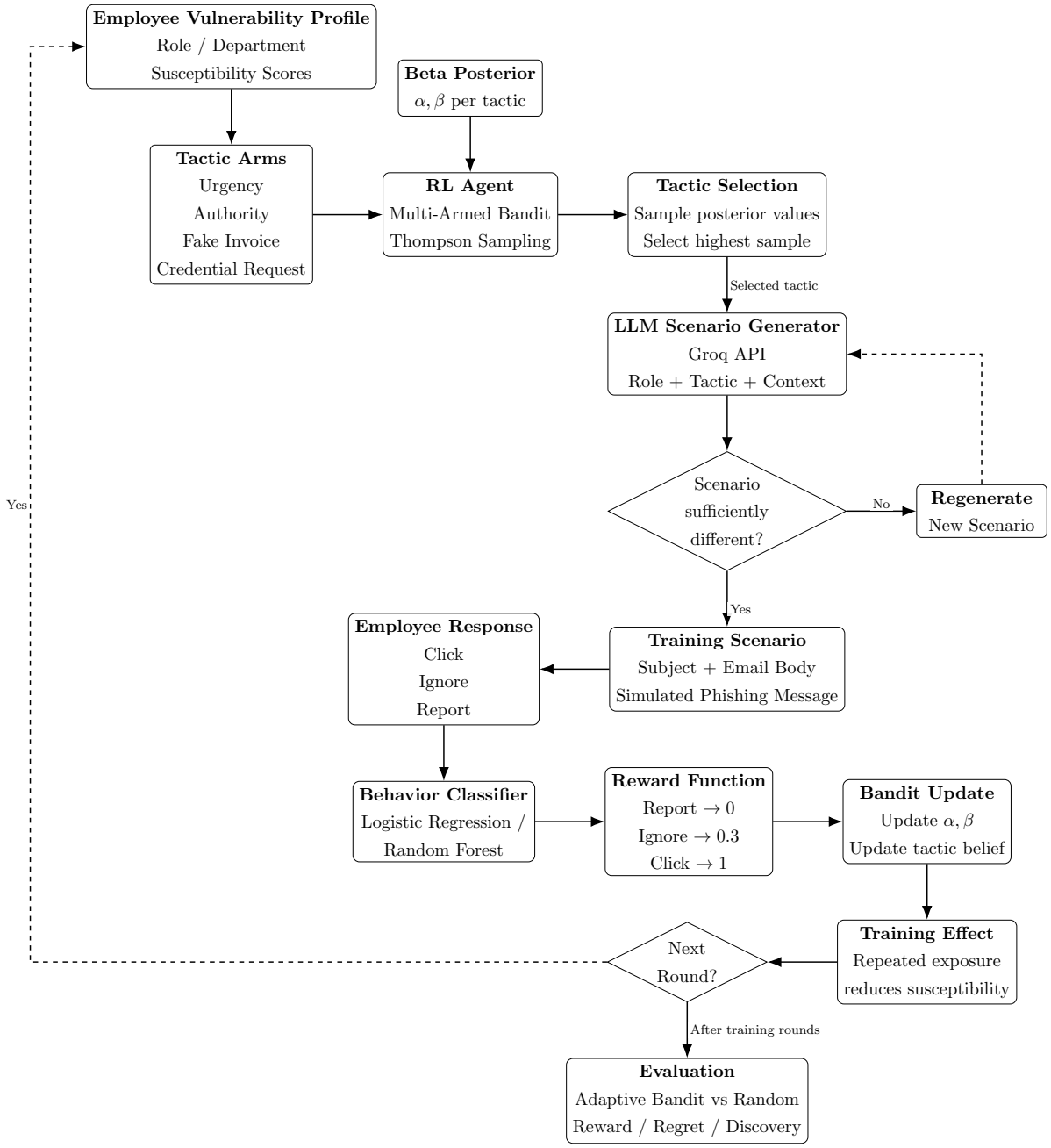


Figure 1: Detailed design methodology of the proposed adaptive security awareness training system

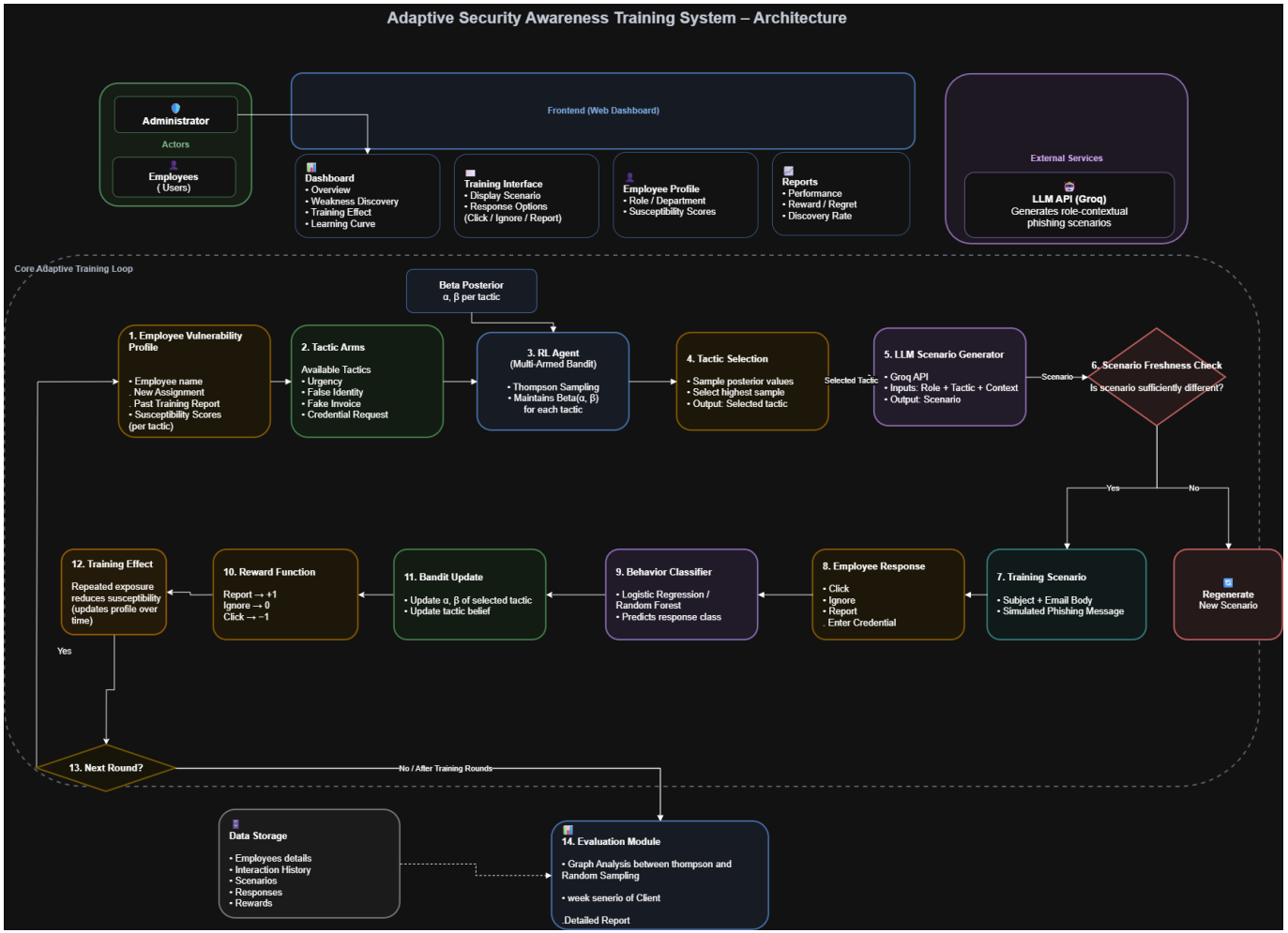


Figure 2: System architecture supporting the adaptive security awareness training methodology

9. Architecture Modules and Description

The architecture is divided into a small set of modules, each with a clear role in the training loop.

9.1 Employee Vulnerability Profile

The employee profile provides the context needed to personalize an exercise. It stores the employee's synthetic role, department, response history, and susceptibility to urgency, authority impersonation, fake invoices, and credential requests. These values are hidden from the bandit during evaluation so that it must learn from observed behavior.

9.2 Reinforcement Learning Agent

The reinforcement learning agent decides which tactic should be shown next. Each tactic is represented as one bandit arm, and Thompson Sampling maintains a probability distribution for every arm. Sampling from these distributions gives less-tested tactics a chance to be explored while still favouring tactics that have produced useful feedback. After each round, the agent updates the distribution of the selected tactic.

9.3 LLM Scenario Generator

The LLM scenario generator turns the selected tactic into a readable, role-appropriate exercise rather than reusing the same fixed email. Its input can include:

- selected social-engineering tactic;
- employee role;
- department;
- scenario context; and
- previous scenario information.

Using this information, the module creates a fictional message for the controlled training environment. The content is clearly treated as a simulation and is never sent as a real phishing email.

9.4 Behavior Classifier

The behavior classifier provides a consistent way to interpret the response produced by the employee simulator. A lightweight model such as logistic regression or random forest can be trained on synthetic, labeled examples, making the prototype practical to run on a standard laptop.

Possible response classes include:

- Click;
- Ignore;
- Report.

This classifier is only a component of the simulation. Its predictions should not be presented as a validated model of real human behavior without evidence from an appropriate human-participant study.

9.5 Reward Function

The reward function translates the classified response into a value the learning agent can use. For vulnerability discovery, a risky response produces a stronger internal signal:

$$R(\text{report}) = 0$$

$$R(\text{ignore}) = 0.3$$

$$R(\text{click}) = 1$$

This internal reward helps the agent locate tactics that deserve more attention. It is separate from the learner-facing safety score, where reporting a suspicious message is the preferred outcome. Keeping the two measures distinct avoids rewarding unsafe behavior in the training interface.

9.6 Training Loop

A complete round therefore follows these steps:

1. Select a social-engineering tactic.
2. Generate a contextualized scenario.
3. Simulate the employee response.
4. Classify the response.
5. Calculate the reward.
6. Update the bandit.
7. Select the next tactic.

9.7 Dashboard and User Interface

The implemented frontend provides separate workspaces for administrators and employees. Administrators can view the organization risk overview, manage training assignments, inspect model health, and review training analytics. Employees can view their own assignments, personal progress, training feedback, and active simulations.

The dashboard navigation was kept intentionally simple by removing the unused Manage, Employees, Learning Path, Settings, and Preferences sections. The remaining workspace links focus on the screens used by the prototype. Risk graphs are shown in both workspaces with clearly labelled axes: the horizontal axis represents training progress over time, while the vertical axis represents the risk score from 0% to 100%. A lower line indicates lower risk. The administrator view also includes a grey comparison line representing performance without targeted training.

10. Dataset Description

The prototype does not depend on an external public dataset. Instead, it uses configurable synthetic employee profiles and records the interactions generated during each experiment. Every simulated employee is assigned a hidden susceptibility value for each social-engineering tactic.

For example, a simplified profile may look like the following:

Table 2: Example Simulated Employee Vulnerability Profile

Tactic	**Susceptibility**
Urgency	0.30
Authority impersonation	0.45
Fake invoice	0.80
Credential request	0.25

Scenario-response examples for the behavior classifier are generated from these profiles. A single training record can contain:

- employee identifier;
- employee role;
- selected tactic;
- generated scenario;
- scenario features;
- simulated response;
- reward;
- training round.

Because the underlying susceptibility values are known, the synthetic dataset makes it possible to check whether the bandit identifies the intended weakness. At the same time, the results must be interpreted as an algorithmic evaluation rather than evidence about real employees.

11. Results and Analysis

The implemented prototype successfully connects the synthetic employee profiles, Thompson Sampling policy, scenario generation, response classification, and dashboard visualization into one training loop. Each completed round records the selected tactic, scenario, response, reward, and training progress. This allows the system to analyse both individual performance and organization-level risk.

The administrator dashboard summarizes the number of employees, rounds analysed, average reporting time, and repeat clickers. It also shows the risk trend across training rounds and provides model-health information. The employee dashboard presents assignments, personal risk progress, feedback, and the next available simulation. In both dashboards, a lower risk-score line represents safer performance over time.

The current results should be treated as prototype results rather than general claims about human behaviour. The data is synthetic and is used to verify that the software records interactions, updates the policy, and presents the resulting trends correctly.

11.1 Administrator Console

The administrator console provides an organization-level view of the training system. As shown in Figure 3, it presents the overall risk score, number of employees, analysed rounds, average reporting time, repeat clickers, and the risk trend across training rounds. This enables an administrator to monitor whether targeted training is reducing risk and to identify users who may require additional practice.

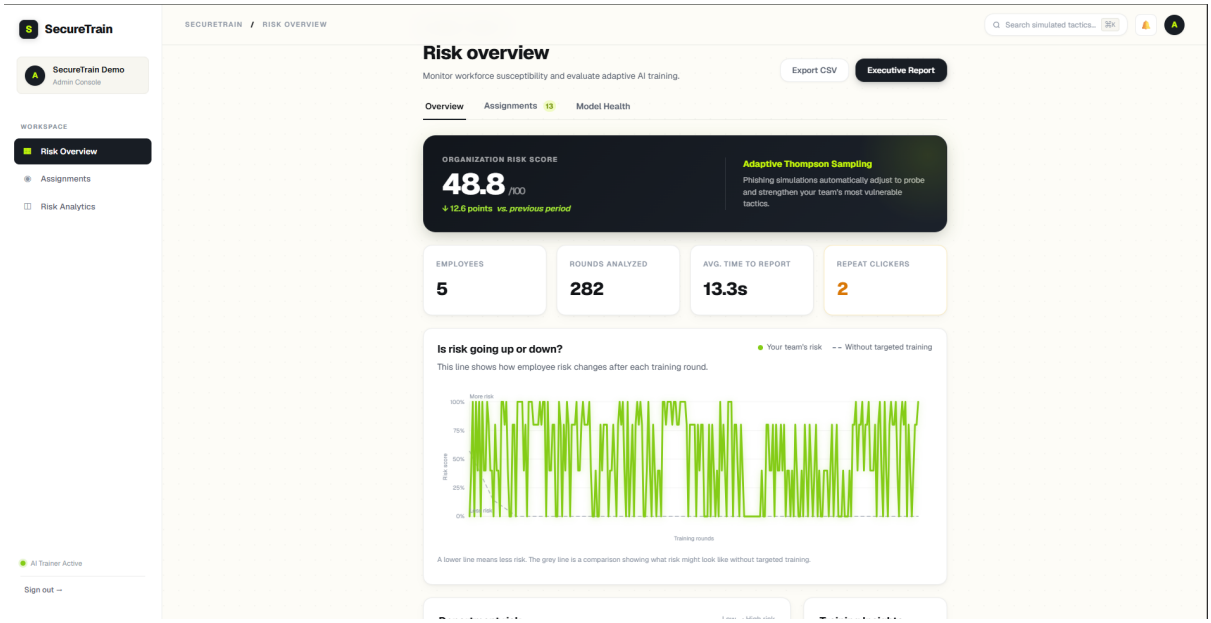


Figure 3: Administrator console showing organization-level risk and training statistics

11.2 Employee Console

The employee console focuses on individual progress. Figure 4 shows the employee's personal risk score, risk trend, strongest tactic, current practice target, and recent responses. These indicators give the learner direct feedback about areas of strength and the tactic that should receive greater attention in later rounds.

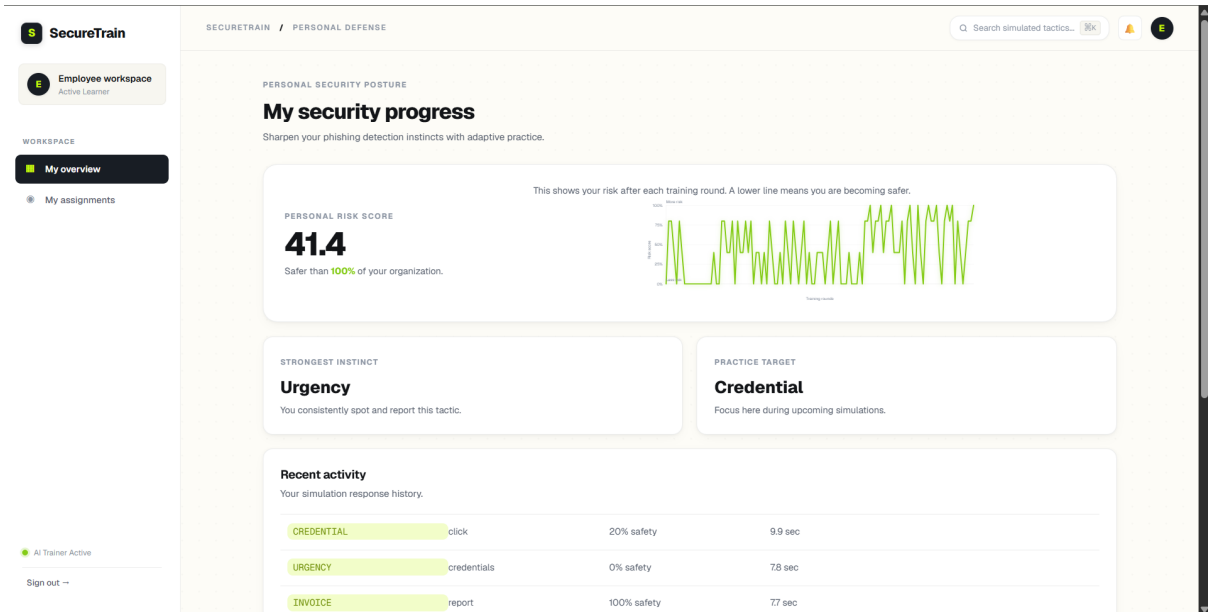


Figure 4: Employee console showing personal security progress and recent activity

11.3 Training Assignment Creation

Administrators create a training assignment by selecting an employee, entering an assignment title, and specifying the number of rounds. Figure 5 also shows the current assignments and their completion status, allowing completed campaigns to be opened for further analysis.

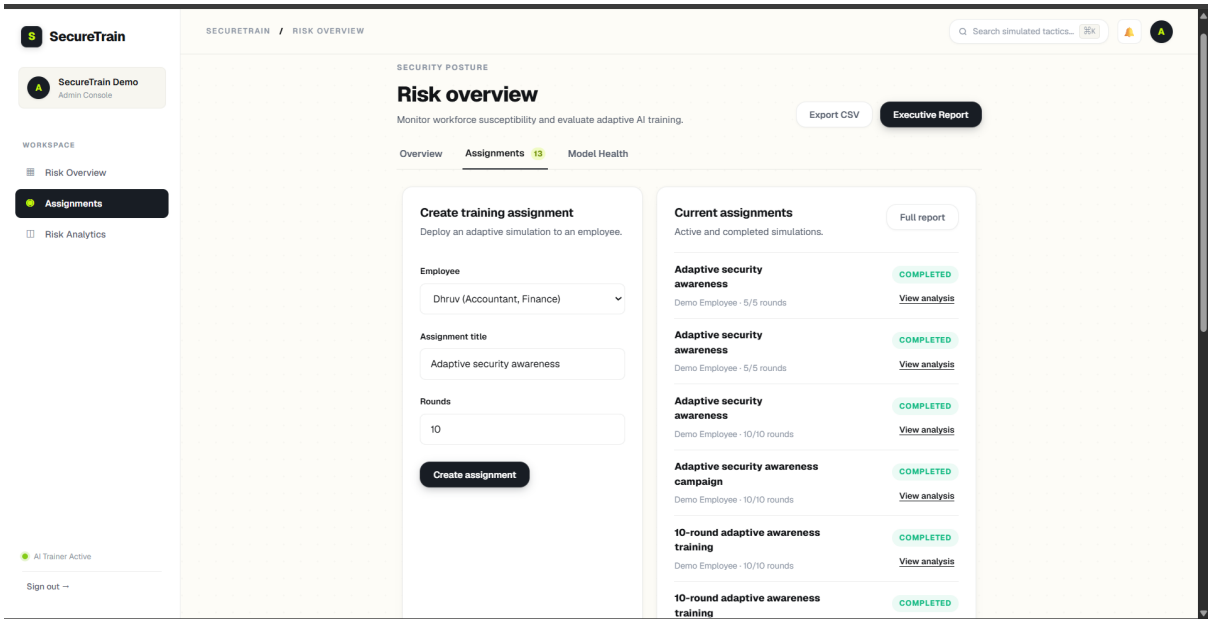


Figure 5: Administrator interface for creating and monitoring training assignments

11.4 Employee Training Rounds

During each round, the employee receives a simulated phishing message generated for the selected tactic. Figure 6 shows an invoice-based scenario with the available responses: report phishing, ignore, open the link, or submit credentials. The selected response, response time, tactic, and safety score are stored before the bandit selects the tactic for the next round.

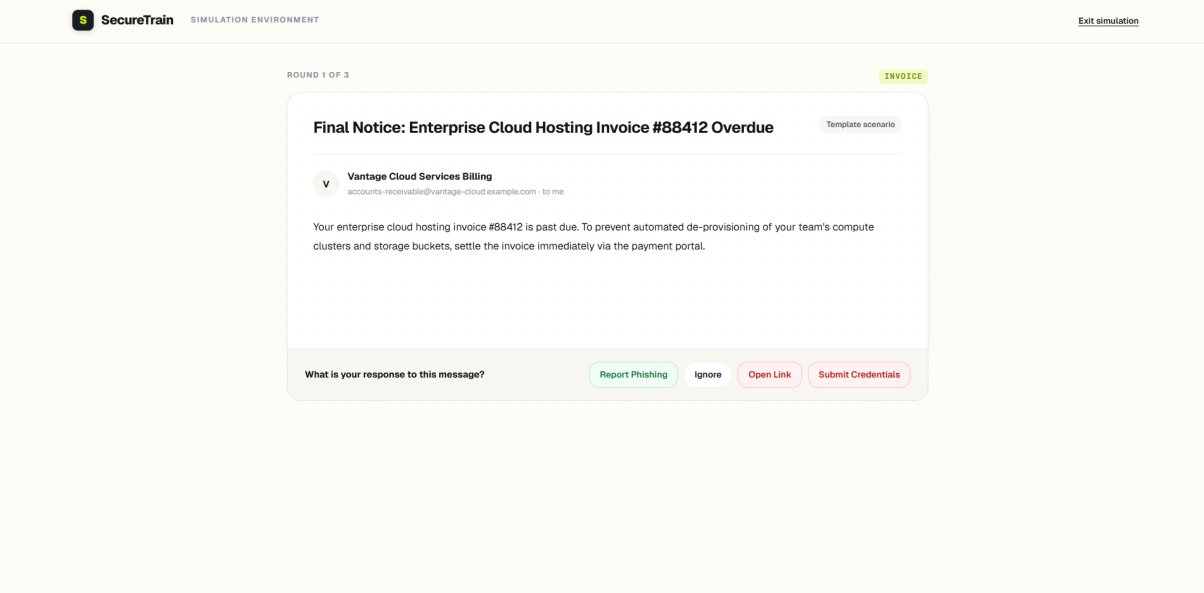


Figure 6: Employee completing an adaptive phishing-simulation round

11.5 Round and Assignment Analysis

After the assignment is completed, the system analyses the records from all individual rounds. The report in Figure 7 summarizes average safety, detection reward, the weakest tactic, and performance by tactic. It also generates a coaching summary from the employee's round-by-round responses, explaining unsafe actions and recommending the behaviours that should be practised in future rounds.

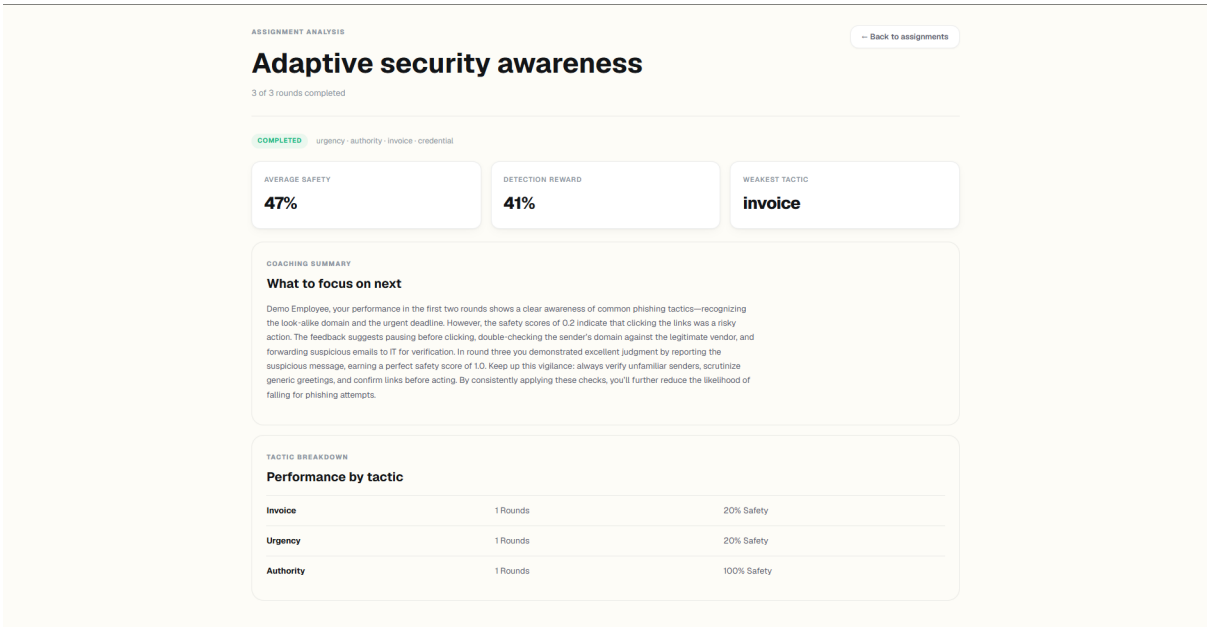


Figure 7: Detailed analysis generated from the completed training rounds

12. Expected Outcome

The expected outcome is that the adaptive policy will identify tactics that produce weaker responses and select them more often for targeted practice. With repeated rounds, employees should receive more relevant scenarios instead of following the same fixed sequence. The system is also expected to generate varied, role-aware scenarios while keeping the training process controlled and measurable.

Success will be assessed using the recorded reward, risk score, reporting behaviour, tactic-selection frequency, and trend across training rounds. The dashboard should make these changes understandable to users without requiring them to interpret the internal bandit parameters.

13. Comparison Analysis

The adaptive system is compared with a random tactic-selection baseline. Both approaches use the same synthetic employee profiles, number of training rounds, scenario-response process, and reward calculation. The only difference is how the next tactic is selected.

Table 3: Adaptive Policy and Random Baseline Comparison

Measure	**Adaptive Thompson Sampling**	**Random selection**
Tactic selection	Uses previous responses and rewards	Selects tactics without learning
Personalization	Gives more attention to observed weak areas	Gives similar exposure to all tactics
Exploration	Balances exploration and targeted practice	Has no policy-based balance
Scenario focus	Can concentrate on tactics needing more training	May repeat strong or weak tactics unpredictably
Expected learning value	More focused training over time	Useful as a simple reference point

The adaptive approach is expected to achieve a more informative reward trend and faster discovery of the intended weakness than random selection. The comparison does not prove that one method is better for real employees; it only measures the behaviour of the two policies in the controlled synthetic environment.

14. Conclusion

This project brings adaptive tactic selection, LLM-based scenario generation, synthetic employee modeling, behavior classification, and response-based feedback into one working training loop. Thompson Sampling offers a transparent way to balance the exploration of less tested tactics with the use of tactics that have already produced useful evidence. The synthetic environment also provides a fair basis for comparing this adaptive approach with random selection.

The implemented dashboard also provides separate administrator and employee views with simplified navigation and clearer risk visualizations. The graphs label the training timeline and risk-score axis so that users can understand whether risk is increasing or decreasing without needing to interpret technical model details.

The result is intended to be a practical proof of concept rather than a validated model of employee psychology. It demonstrates how security awareness training could respond to individual behavior while remaining within a controlled, consent-based setting. No real phishing campaign is conducted, and no real credentials are requested or collected.

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